

PARTNERSHIP

1. A person starts a business with Rs.45,000. Later his friend joined him with Rs.20,000 after 4 months. What is the share of A, if their annual profit is Rs.1,40,000?
a) Rs.10,8000 b) Rs.13,8000 c) Rs.32,000
d) Rs.46,000 e) None of these
2. A, B and C enters into a partnership investing Rs.2500, Rs.3500 and Rs.4000. Find their respective shares in the annual profit of Rs.14,700?
a) Rs.3500,Rs.6500,Rs.1500
b) Rs.3500,Rs.4900,Rs.6500
c) Rs.3675,Rs.5145,Rs.5880
d) Rs.3500,Rs.6300,Rs.4900 e) None of these
3. Sanjay started a business by investing Rs.40,000. After 6 months Nirmal joined him with a capital of Rs.60,000. After 3 years, they earned a profit of Rs.27,900, what was sanjay's share in the profit?
a) Rs.12400 b) Rs.13000 c) Rs.13200
d) Rs.15000 e) None of these
4. Sofia started a business investing ₹12000.
At the end of the year, the total profit was ₹4416. What is the share of profit of sofia?
a) ₹1344 b) ₹3364 c) ₹3072 d) ₹2568
5. Rahul started a business investing ₹45000. After 8 months Anisha joined her with a capital of ₹52000. At the end of the year the total profit was ₹56165. What is the share of the profit of Anisha?
a) ₹56000 b) ₹35000 c) ₹72000 d) None of these
6. P, Q and R invested ₹45,000, ₹70,000 and ₹90,000 respectively to start a business. At the end of two years, they earned a profit of ₹1,64,000. What will be the Q's share in the profit?
a) ₹56000 b) ₹35000 c) ₹72000 d) ₹64000
7. Yuvan started a business investing Rs.45,000. After 3 months, Pranav joined him with a capital of Rs.60,000. After another 3 months, Ayan joined them with a capital of Rs.90,000. At the end of the year, they made a profit of Rs.20,000. What would be the share of Ayan?
a) Rs.7666.66 b) Rs.6666.66 c) Rs.5666.66
d) Rs.4666.66 e) None of these
8. Janani started a business investing Rs.40,000. After 6 months, Pooja joined her with a capital of 55,000. What is the difference in their respective shares if the annual profit is Rs.1,89,000?
a) Rs.24,000 b) Rs.35,000 c) Rs.20,000
d) Rs.19,000 e) None of these
9. A, B and C rent a system. A types 10 programs in 7 days, B types 12 programs in 5 days and C types 15 programs in 3 days. If the rent of the system is Rs.175000, What is the difference between the rental shares of B and C?
a) Rs.64,000 b) Rs.51,000 c) Rs.15,000
d) Rs.10,000 e) None of these

10. Praveen and Robin started a business investing ₹42,000 and ₹57,000 respectively, but Robin left the business after 8 months. Find the difference between the share of profits of Praveen and Robin, if the total profit at the end of the year was ₹26000?

- a) ₹1200 b) ₹1400 c) ₹1600 d) ₹1300

11. Prabha started a business investing 32000. After 5 months, Amit joined her with a capital of ₹22,000. At the end of the year the total profit was ₹16409. What is the difference between the shares of prabha and amit?

- a) ₹5284 b) ₹11712 c) ₹10182 d) ₹7015

12. Aurn started a business investing ₹3800. After 5 months Bakul joined him with a capital of ₹55,000. At the end of the year the total profit was ₹22,000. What is the approximate difference between the share of profit of Arun and Bakul?

- a) ₹1192 b) ₹1856 c) ₹1007 d) ₹1928

13. A and B started a business investing Rs.16,000 and Rs.12,000 respectively. After 3 months, A withdraws Rs.5000 while B invested Rs.5000 more. After 3 more months, C joins the business with a capital of Rs.21,000. The share of B exceeds that of C by, out of a total profit of Rs.26400 after one year by?

- a) Rs.2400 b) Rs.3600 c) Rs.3000
d) Rs.4800 e) None of these

14. A, B and C started a business by investing Rs.42,000, Rs.30,000 and Rs.28,000 respectively. After 4 months, A withdraws Rs.12,000, B withdraws Rs.6000 and C withdraws Rs.8000. If after 10 months, a total profit earned is Rs.46420. What is 20% of the share of C?

- a) Rs.2563 b) Rs.2552 c) Rs.2673
d) Rs.2555 e) None of these

15. Sarita started a business investing ₹50000. After six months, Abhishek joined her with ₹75000. After another six months, Nisha also joined them with ₹1.25 lakh. What should be the ratio in which the profits are shared by the 3 persons respectively at the end of 2 years?

- a) 4:5:6 b) 8:9:10 c) 8:9:12 d) 4:5:8

ASSIGNMENT

16. Subhash started a business by investing ₹25,000. 6months later Aditya joins him by investing ₹15,000. After another 6 months. Aditya invests an additional amount for ₹15,000. At the end of 3 years they earn a profit of ₹247000. What is the aditya's share in the profit?

- a) ₹1,13,000 b) ₹1,23,000
c) ₹1,17,000 d) ₹1,11,500

17. Prakash, sunil and Anil started a business jointly investing ₹11 lakhs, ₹16.5 lakhs and ₹8.25 lakhs. What will be the 50% of Anil's share in the profit?

- a) ₹4.5 lakhs b) ₹2.25 lakhs
c) ₹2.5 lakhs d) cannot be determined

18. A, B and C enter into a partnership. A contributes ₹320 for 4 months, B contributes ₹510 for 3 months and C contributes ₹270 for 5 months. If the total profit is ₹208, what will be the share of A?

- a) ₹92 b) ₹64 c) ₹88 d) ₹92

19. Sreeja started a business with Rs.45000. Later, Madhu joined her with Rs.54000. When did Madhu join if the profits at the end of the year were divided in the ratio of 2 : 1?

- a) 4th month b) 5th month c) 7th month
d) 6th month e) None of these

20. Anitha and suji are partners in a business. Anitha contributes $\frac{1}{4}$ of her capital for 15 months and suji received $\frac{2}{3}$ of the profit. For how long suji's money was invested in the business?

- a) 1 year b) 6 months c) 10 months
d) 9 months e) None of these

21. Ajay and Arjun enters into a partnership with their respective capitals in the ratio 5 : 6. At the end of 8 months, Ajay withdraws his capital. If they receive their shares of profit in the ratio 5 : 9, find how long Arjun's capital was invested?

- a) 10 months b) 12 months c) 14 months
d) 15 months e) None of these

22. Ravi and Kumar invests in a business in the ratio of 3 : 2. If 5% of the total profit goes to charity and Ravi's share is Rs.855, what is the total profit?

- a) Rs.1576 b) Rs.1576.50 c) Rs.1425
d) Rs.1537 e) None of these

PERCENTAGE

Percent means per hundred. $r\%$ means $\frac{r}{100}$

A percent can be expressed as a fraction with a denominator of 100 or as a decimal.

For example, $19\% = \frac{19}{100} = 0.19$

To find a certain percent of a numbers, multiply the number by the percent expressed as a fraction or as a decimal.

For example: $5\% \text{ of } 300 = \frac{5}{100} \times 300 = 15$

Percent change.

Increase Percentage = $\frac{\text{Actual Decrease}}{\text{Original Value}} \times 100$

For example, if the price of an item increases from 25 to 30, increase percentage is given by,

Increase percentage = $\frac{5}{25} \times 100 = 20\%$

Decrease Percentage = $\frac{\text{Actual Decrease}}{\text{Original Value}} \times 100$

For example, if the price of an item is reduced from 30 to 25,

Decrease percentage = $\frac{5}{30} \times 100 = 16 \frac{2}{3} \% = 16.67 \%$

We observe that percentage increases from 25 to 30 is not the same as percentage decrease from 30 to 25.

Some important results

1.If a number is increased by $x\%$ and further increased by $y\%$, then

net percentage increase = $\left(x + y + \frac{xy}{100}\right)\%$

2.If a number is increased by $x\%$ and then decreased by $y\%$, then there will be increase or decrease depending on the value obtained is positive or negative.

decrease percentage = $\left(x - y - \frac{xy}{100}\right)\%$

3.If a number is decreased by $x\%$ and further decreased by $y\%$, then *net percentage decrease* $= \left(-x - y + \frac{xy}{100}\right)\%$

4.If a number A is $x\%$ more than B, then B is $\left[\frac{x}{100+x} \times 100\right]\%$ less than A.

5.If a number A is $x\%$ less than B, then B is $\left[\frac{x}{100-x} \times 100\right]\%$ greater than A.

6.If a number A is increased by $x\%$, the number B should be reduced by $\left[\frac{x}{100+x} \times 100\right]\%$ so as to keep the product AB of the numbers, the same.

If a number A is decreased by $x\%$, then the number B should be increased by $\left[\frac{x}{100-x} \times 100\right]\%$ so as to keep the product AB of the numbers, the same.

Example.

1. What percent of 350 is 6?

A) 21% B) 1.7% C) 1% D) 2% E) 35%

If $x\%$ of 350 is 6, then $\frac{x}{100} \times 350 = 6 \Rightarrow x = \frac{12}{7} = 1\frac{5}{7}\% = 1.7\%$

$\therefore 1.7\%$ of 350 is 6 approximately (B)

2. If the price of petrol is increased by 25% how much percent a motorist must reduce his consumption of petrol so as not to increase or decrease his expenditure on petrol.

A) 25 % B) 20 % C) 16 2/3 %

D) 12 1/2 % E) 17 %

If price of petrol is increased by 25% consumption should be reduced by $\frac{25}{100+25} \times 100 = \frac{25}{125} \times 100 = 20\%$. (B).

Consumption should be reduced by 20% so as to keep the same expenditure.

EXERCISE

1. 35% of a number is 182. What is 130% of that number?

a) 676 b) 667 c) 673 d) 663 e) 656

2.If 40% of a number is 256. Then what is 25% of that number?

a) 140 b) 160 c) 190 d) 220 e) 230

3. 40% of 70% of a number is 336. What is one- third of that number?

a) 475 b) 480 c) 420 d) 450 e) 400

4. The difference between 31% of a number and 13% of the same number is 576. What is 17% of that number?

a) 544 b) 457 c) 561 d) 574 e) 554

5. $\frac{3}{5}$ th of a number is 250 more than 40% of the same number. What is number?

a) 1250 b) 1180 c) 1200 d) 1220 e) 1300

6. Jyoti's monthly expense on shopping is 25%, travelling is 20%, food is 45%, and 10% on other recreational activities. What is the total amount she spent on travelling and food together?

a) Rs.15000 b) Rs.18000 c) Rs.12050
d) Cannot be determined e) Rs.14050

7. Anish spends 25% of a salary on house rent 5% on food, 15% on travel, 10% on clothes and the remaining amount of Rs.22500 is saved. What is Anish salary?

- a) Rs.40000 b) Rs.40500 c) Rs.45500
d) Rs.50000 e) Rs.45000

8. It is required to get 40% of marks to pass an exam. A candidates score 200 marks and failed by 8 marks. What were the maximum marks of that exam?

- a)530 b)540 c)502 d) Cannot be determined e) 520

9. 1100 boys and 700 girls are examine in a test; 42% of the boys and 30% of the girls pass. The percentage of the total who failed is?

- a) 58% b) $62\frac{2}{3}\%$ c) 64% d) 78% e) $72\frac{2}{3}\%$

10. The price of a car is Rs 500000/- It was insured for 90% of its price. The car got completely damaged in an accident and the insurance company paid by only 80% of the insured amount. What is the difference between price of the car and amount of insurance received?

- a)150000 b)160000 c)140000 d)130000

11. Naveen decided to donate 15% of his salary to a charity. On the day of donation he changed his mind and donated Rs.1896, which was 80% of what he had decided earlier, How much is Naveen's salary?

- a) 18500 b) 10250 c) 15800 d) None of these

12. In an election between two candidate the winner secured 58% of the total votes cast and wins by a majority of 2400 votes. How many votes did the loser get?

- a) 6100 b) 6200 c) 6300 d) 6400

13. Raman's salary was decreased by 50% and subsequently increased by 50%. How much percent does he lose?

- a) 75 b) 25 c) 50 d) 100

14. A number is increased by 20% and then again by 20%. By what percent should the increased number be reduced so as to get back the original number?

- a) $19\frac{11}{31}\%$ b) $30\frac{5}{9}\%$ c) 40% d) 44%

15. Peter earned 40% more money than Albert. Albert earned 20% less than Michael. Peter earned more than Michael by:

- a) 10% b) 12% c) 20% d) 25%

ASSIGNMENT

16. 61% of a number is 9028. What is 46% of that number?

- a) 6660 b) 6956 c) 7104 d) 6808 e) 6608

17. Five-ninth of sixty two percent of a number is equal to 2790. What is the number?

- a) 8100 b) 7200 c) 6300 d) 8500 e) 7900

18. 56% of a number is 463.68. What is 25% of that number?

- a) 202 b) 204 c) 206 d) 208 e) 207

19. 25% of 50% of a number is 385. What is the number?

- a) 3050 b) 3100 c) 3004 d) 3080 e) 3060

20. What is the value of three fourth of sixty percent of 480?
a) 216 b) 218 c) 212 d) 214 e) 210
21. 37% of a number is 114.7. What is 130% of that number?
a) 406 b) 405 c) 403 d) 407 e) 400
22. 43% of a number is 559. What is 32% of that number?
a) 416 b) 429 c) 403 d) 384 e) 614
23. Twenty – Seven percent of Arun’s monthly salary is equal to Raju’s monthly salary. If Arun’s monthly salary is Rs.25000, then what is Raju’s annual salary?
a) Rs.87000 b) Rs.810000 c) Rs.111000
d) Rs.57000 e) Rs.81000
24. Manjeet spent 12% on bills, 22% of rent and 28% on shopping $\frac{1}{4}$ th of the remaining amount is Rs.3610. How much did he spend on bills?
a) Rs.4580 b) Rs.4530 c) Rs.4520
d) Rs.4560 e) Rs.4510
25. There are 3050 employees in an organizations. Out of which 48% got transferred to different places. How many employees did not got transferred?
a) 1384 b) 1628 c) 1464 d) 1526 e) 1586

PROFIT AND LOSS

When goods are purchased for one sum of money and sold for another, there is a profit or loss, according as the selling price is greater or less than the cost price.

Profit = Selling price - Cost price

Loss = Cost price - Selling price

Percentage of profit or loss is generally represented as a percentage of cost price, unless, otherwise specified.

$$\text{Profit Percentage} = \frac{\text{Actual Profit}}{\text{Cost Price}} \times 100$$

$$\text{Loss Percentage} = \frac{\text{Actual Loss}}{\text{Cost Price}} \times 100$$

$$\text{Cost Price} = \frac{\text{Selling Price} \times 100}{100 + \text{Profit Percent}} = \frac{\text{Selling Price} \times 100}{100 - \text{Loss Percent}}$$

[When there is a Loss] [When there is a Loss]

$$\text{Selling Price} = \frac{(\text{Cost Price})(100 + \text{Profit Percent})}{100} = \frac{(\text{Cost Price})(100 - \text{Loss Percent})}{100}$$

When selling Price of two articles is the same, one is sold at a gain of $r\%$ and the other is sold at a loss of $r\%$, there is always a loss and $\text{Loss Percentage} = \frac{r^2}{100}\%$

DISCOUNT.

Some low priced article say such as shirt, trousers, saris or salwar has a tag which indicates the price at which it is sold. Some of the high priced articles such as four wheelers and two wheelers has a list on which the price is indicated. Such tag price or list price is called MARKED PRICE (MP). Discount is calculated on the marked price.

Suppose the marked price of an article is Rs. 400 and the discount offered is 20%, then the discount = $\frac{20}{100} \times 400 = 80$. Hence the selling price = $400 - 80 = 320$.

$$\text{Discount in rupees} = \text{mp} - \text{sp} \quad d\% = \frac{\text{mp} - \text{sp}}{\text{mp}} \times 100 \quad \text{sp} = \text{mp} \frac{100 - d\%}{100} \quad \text{mp} = \text{sp} \frac{100}{100 - d\%}$$

Net Selling Price = M.P $\left(1 - \frac{d1\%}{100}\right) \left(1 - \frac{d2\%}{100}\right)$ after a series of two discount of d1 % and d2 %.
 Two Successive discounts of d1% and d2 % are equivalent to a single discount of $\left(d1 + d2 - \frac{d1 \times d2}{100}\right)\%$

Examples

1. After getting two successive discounts of 20 % and 10 % Judy paid \$ 360 for a bicycle. How much in dollars, did the bicycle originally cost?

- A) \$ 400 B) \$ 510 C) \$ 500
 D) \$ 440 E) \$ 514

Two discounts of 20% and 10% are equivalent to a single discount of

$$20 + 10 - \frac{20 \times 10}{100} \%$$

$$\text{M.P} = \$ 500 \text{ (C)}$$

2. By selling a table for \$ 162, sherry loses 10%. For how much should he sell the same to gain 20%.

- A) \$ 177.96 B) \$ 180 C) \$ 200
 D) \$ 204 E) \$ 216

$$\text{S.P} = \$ 162$$

$$\text{C.P} = \frac{162 \times 100}{90} = \$ 180$$

New Selling Price = \$ 180 + 20% of 180 = \$ 216 (E)

3. A trader allows a discount of 5% for cash payment. How much percent above the cost piece must he mark to make a profit of 14%.

- A) 19 % B) 9 % C) 20 %
 D) 24 % E) 25 %

Let the CP be \$ 100. Selling Price = \$ 114

$$114 = \text{MP} \left(1 - \frac{5}{100}\right). \text{MP} = \$ 120.$$

∴ 20% above the cost price must be marked.

EXERCISE

1. An umbrella was sold at a profit of 20%. What is the selling price of the umbrella, If its cost price is Rs 180?

- a)280 b)200 c)216 d)160

2. An article was sold for Rs 13000 at a loss of 35% what is the cost price of the article?

- a)1500 b)1800 c)2200 d)2000

3. A person incurs 5% loss by selling a watch for Rs 1140. At what price should the watch be sold to earn 5 % profit?

- a) Rs 1280 b) Rs 1260 c) Rs 1360 d) Rs 1180

4. The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is:

- a) 15 b) 16 c) 18 d) 25

5. A man sells a plot of land at 6% profit. If he had sold it at 10% profit, he would have received Rs. 200 more. What is the selling price of the land?
a) Rs. 5000 b) Rs. 5300 c) Rs. 4800 d) Rs. 5500
6. A makes an article for Rs. 120 and sells it to B at a profit of 25%. B sells it to C who sells it for Rs. 198, making a profit of 10%. What profit percent did B make?
a) 25% b) 20% c) 16.66% d) 15%
7. A watch was sold at a loss of 9%. It was observed that if the selling price was Rs 420 more than profit made would have been 5%. What was the actual price of the watch?
a) 2730 b) 2780 c) 3100 d) 3150
8. Praveen sold an article for Rs 1170 at a profit of 30%. What should be the selling price of the desired profit is 40%?
a) 1240 b) 1250 c) 1250 d) 1260
9. The profit earned by selling a phone for Rs 18000 is same as the loss incurred after selling for Rs 16800. What is the cost price of the phone?
a) 25000 b) 20000 c) 17400 d) 1850
10. The profit earned by selling a shirt for Rs 1200 is twice the loss incurred when the shirt is sold for Rs 600. What is the cost price of the shirt?
a) 800 b) 700 c) 600 d) 500
11. A trader marks his goods at 40% above the cost price and allows a discount of 25%. What is his gain percent?
a) 3% b) 7% c) 6% d) 5%
12. Find the single discount equivalent to two successive discounts of 20% and 10%
a) 30% b) 28% c) 10% d) 25%
13. A shopkeeper marks his goods in such a way that even after allowing a discount of 20%. He marks a profit of 12%. How much percent above the cost price is the marked price?
a) 32% b) 8% c) 12% d) 40%
14. Sridhar sold 16 pens at the cost of 20 pens. What is the profit or loss percentage made by him?
a) 25% b) 20 c) 50% d) 35%
15. A shopkeeper bought two ceiling fans for Rs 800 each. He sold one fan at a profit of 12% and the other at a loss of 12%. What would be his overall profit or loss in the transaction?
a) 20% profit b) 20% loss
c) No profit, no loss d) cannot be determined

ASSIGNMENT

16. A sold an article to B to a profit of 20% and B sold the same article to C at a profit of 10% . if C bought it for Rs 2640 . How much did A pay for it?
a) 1500 b) 2000 c) 2500 d) 3000
17. A Dishonest dealer claims to sell his goods at the cost price but uses a weight of 800gm instead of 1kg. What will be the profit percentage in this transaction?
a) 20% b) 25% c) 30% d) 35%

18. If the cost price is 96% of the selling price, then what is the profit percentage?

- a) 4.11% b) 4.15% c) 4.17 d) 4.20%

19. By selling an article for Rs.100, a man gains Rs.15. Then, his gain% is:

- a) 15% b) $12\frac{2}{3}\%$ c) $17\frac{11}{17}\%$ d) $17\frac{1}{4}\%$

20. Ram bought a scooter for a certain sum of money. He spends 10% of the cost on repairs and sold the scooter for a profit of Rs 1100. How much did he spend on repair if he made a profit of 20%?

- a) Rs 450 b) Rs 500 c) Rs 600 d) Rs 700

21. If a man reduces the selling price of a fan from Rs.400 to RS.380. His loss increases by 2%. The cost price of the fan is

- a) 480 b) 500 c) 600 d) none of these

22. The shopkeeper fixes the market price of an item 35% above its cost price. The % of discount allowed to gain 8% is

- a) 20% b) 27% c) 31% d) 43%

23. By selling an umbrella for Rs.300, a shopkeeper gains 20%. During a clearance sale, the shopkeeper allows a discount of 10% on the marked price. His gain percent during the sale is:

- a) 7 b) 7.5 c) 8 d) 9

24. Prem sold 10 pens at the cost of 12 similar pens. What % profit or loss does he make in this transaction?

- a) 20% profit b) 25% profit c) 16.66% loss
d) 20% loss e) None of these

25. A dishonest dealer claims to sell his goods at the cost price but uses a weight of 800gm instead of 1 kg. what will be the profit percentage in this transaction?

- a) 15% b) 20% c) 25% d) 32%